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Design and analysis of BLDC motor speed control for electric vehicles powered by solar PV and grid supply

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Abstract

This paper explores the design, analysis, and comparison of different control strategies for managing the speed of brushless direct current (BLDC) motors in electric vehicles (EVs) powered by both solar photovoltaic (PV) systems and grid supply. The proposed system utilizes a dual-source power configuration, where solar energy is optimized using an advanced maximum power point tracking (MPPT) technique via a boost converter. In contrast, grid power is efficiently controlled through a feedforward decoupling method. Various motor control techniques are evaluated, including hysteresis current control (HCC), field-oriented control (FOC), and hybrid approaches such as FOC combined with HCC (FOC+HCC) and an enhanced strategy integrating FOC, HCC, and sliding-mode control (FOC+HCC+SMC). The FOC+HCC+SMC method delivers the best results by significantly reducing torque ripple, enhancing dynamic response, and ensuring precise speed tracking under different load and power conditions. This study provides valuable insights into intelligent motor control and energy management, contributing to more efficient and sustainable EV technology.

Keywords BLDC motor, Speed control, Torque ripple, Hysteresis current control (HCC), Field-oriented control (FOC), Hybrid control (FOC+HCC) & (FOC+HCC+SMC), Feed-forward decoupling control, Robust MPPT

1 Introduction

Over the past decade, environmental pollution has risen significantly due to the emission of greenhouse gases, including carbon dioxide, nitrous oxide, methane, sulfur dioxide, and hydrocarbons. This surge in pollution has contributed to a growing environmental crisis. At the same time, the increasing demand for fuel has led to a continuous rise in fuel prices, making it an increasingly costly commodity. Vehicle engines are one of the major contributors to greenhouse gas emissions, responsible for about 30 to 50% of CO₂ emissions from road transport. The rise in vehicle emissions worldwide is mainly due to the growing population, economic growth, increased vehicle usage, and urbanization. These factors have led to more vehicles on the road, further increasing emissions. Many



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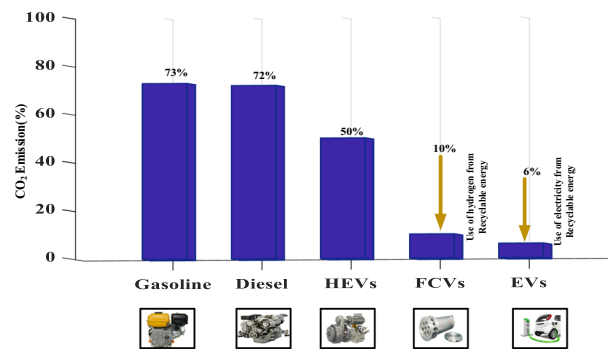


Fig. 1 CO₂ emission levels from vehicles

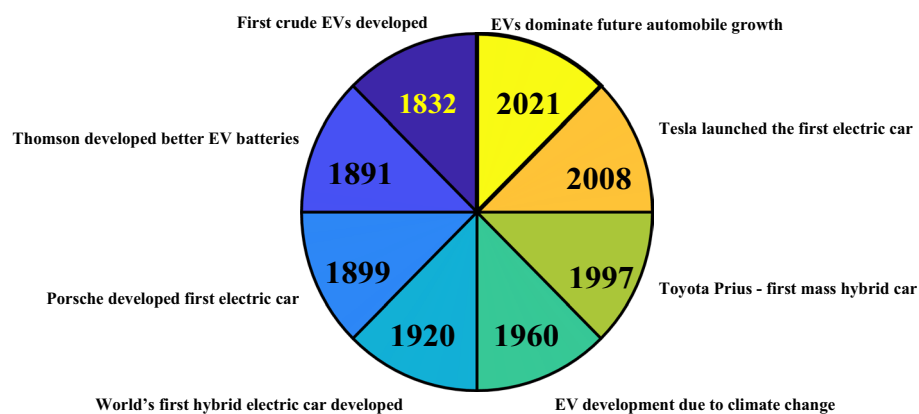


Fig. 2 Timeline of key EV milestones

countries worldwide have begun focusing on reducing harmful gas emissions in the atmosphere and improving air quality everywhere. Electric vehicles produce zero emissions compared to cars that run on internal combustion engines (ICEVs). EVs are much more efficient and hold great potential for reducing CO₂ emissions, especially when powered by cleaner energy sources, as shown in Fig. 1 [1].

The evolution of electric vehicles (EVs) over the past 150 years has been remarkable. From their origin as a basic non-rechargeable design to the advanced modern system we see today, the journey can be divided into 3-distinct phases: the early, middle, and present. These stages reflect the continuous progress in EV technology, as illustrated in Fig. 2 [2].

As highlighted in [3], e-motor technology has advantages and limitations when used to drive an electric vehicle. Over the past century, the automotive industry has made significant changes to motor designs and control methods to enhance the performance of electric vehicles. The main component of every EV is its motor, which directly impacts its efficiency performance, better noise and harness vibration (NHV), high overload capacity, high reliability, high torque, and power density, reasonable cost, and wide constant power and speed range [4].

Various types of electric motors are extensively used in EVs, including permanent magnet synchronous motors (PMSMs), brushless direct current (BLDC) motors, induction motors, and synchronous reluctance motors. Among the various motor options, BLDC motors are more popular in lightweight EVs due to their favorable attributes, such as

a high torque-to-power ratio, low maintenance requirements, high efficiency, compact design, excellent dynamic performance, and a wide speed control range [3–6]. However, precisely controlling the speed of BLDC motors and torque ripple, especially under load-changing conditions, is a complex challenge that requires various control techniques. An optimized power management system for BLDC-driven electric vehicles using a high-gain interleaved Boost-Cuk converter integrated with the grid. Their approach enhances voltage boosting, improves efficiency, and ensures seamless grid integration for reliable EV operation [7]. In general, there are two main techniques: sensor-based and sensor-less. The sensor-based technique uses three hall sensors located 120 degrees apart. In contrast, the sensor-less technique tracks the shape of the back-emf waveform and detects its zero crossing, considering the overall benefits of the sensor-less technique [8]. However, the key disadvantage of sensor-less control is that rapid variations in load and speed might lead to locking by misinterpreting the driving loop. Thus, sensor-based techniques are used in electric vehicles [9]. An optimized design and modeling of a PV-integrated SEPIC-based four-switch inverter for sensorless PMSBLDC motor control. Their system effectively reduces hardware complexity and switching losses while maintaining efficient power conversion. The proposed design enhances motor performance, offers reliable operation under varying solar conditions, and ensures cost-effective, compact solutions for photovoltaic-driven electric vehicle applications. The BLDC motor is the most suitable option for electric two-wheelers. Its speed control capabilities make it highly suitable for use in two-wheeler applications. The BLDC motor was switched using the hall effect sensors and a trapezoidal drive technique, as shown in Fig. 3. An enhanced Direct Torque Control (DTC) strategy for Doubly Fed Induction Machines (DFIM) using a Genetic Algorithm (GA)-optimized PI controller. This method significantly improves torque and flux response, minimizes ripples, and enhances system dynamic performance. The GA-based tuning ensures optimal controller gains, providing better efficiency and stability under various load and speed conditions [10]. These sensors detect the initial rotor position data and regulate commutation after converting to closed-loop control [11, 12]. An optimized PI control strategy for a PV-powered PMSBLDC motor system using a SEPIC-Zeta converter, achieving improved voltage regulation, enhanced dynamic response, and stable motor operation under varying solar conditions [13]. A variable step size MPPT method for photovoltaic systems using a PID

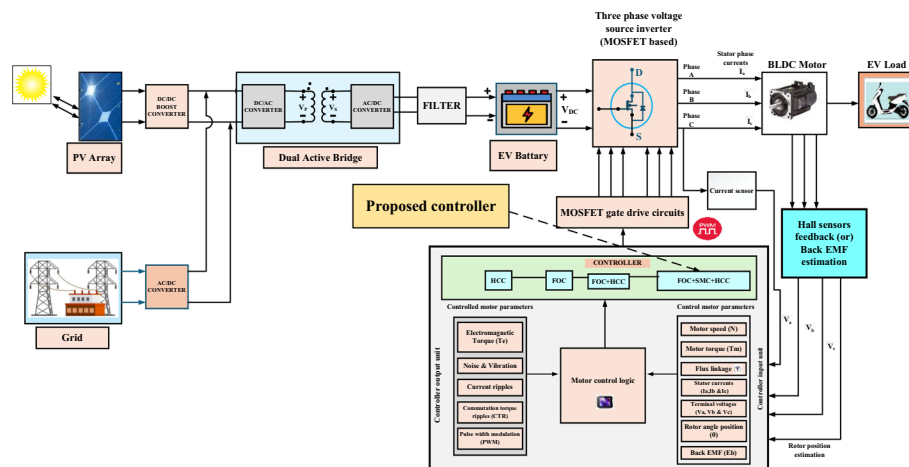


Fig. 3 An overview of solar-grid EV drive

controller optimized by Grey Wolf and Whale Optimization algorithms. This approach enhances tracking accuracy, reduces oscillations, and improves response speed under varying environmental conditions, ensuring efficient solar energy extraction [14].

Traditional HCC ensures a rapid dynamic response but increases switching losses, affecting the system's efficiency [15]. However, a vector control method, i.e., (FOC), a more advanced technique, achieves better efficiency and lower torque ripple by decoupling torque and flux control. Its performance, however, heavily depends on the precise rotor position estimation [16]. To overcome the limitations of individual methods, hybrid control strategies such as the combination of FOC and HCC have emerged, offering a balance between dynamic responses, efficiency, and robustness. However, they have limitations in handling disturbances and ensuring smooth operation under all conditions. To overcome these limitations, this paper introduces the integration of SMC into the FOC+HCC approach. A robust sliding mode control strategy for Z-source inverter-operated BLDC motors in electric vehicles. Their method enhances system stability, reduces sensitivity to disturbances, and improves motor performance, making it suitable for reliable EV propulsion under varying conditions [17]. It addresses these limitations by improving disturbance rejection, making it a more reliable solution for EV applications with varying load conditions. A hybrid MPPT technique combining Artificial Neural Network (ANN) and Grey Wolf Optimization (GWO) algorithms, integrated with Model Predictive Control (MPC) for inverter operation. This approach enhances tracking performance under partial shading conditions and ensures efficient EV charging. Their method demonstrates improved energy extraction, faster convergence, and greater system stability in dynamic solar environments [18]. A 5-kW solar-powered EV charging station using an ANN–Kalman filter MPPT and MPC-based inverter control. The system achieved reduced THD, improved energy efficiency, and reliable performance under varying environmental and load conditions [19]. An efficient single-phase EV charging system combining flyback and buck converters with Model Predictive Control (MPC) to optimize lithium-ion battery charging. The system improves energy efficiency, voltage regulation, and charging performance for enhanced battery management [20]. A PMS motor drive using an ANN-based MPPT controller for solar-battery-powered EVs achieves improved tracking accuracy, enhanced system efficiency, and stable motor performance under varying conditions [21].

This paper compares different speed control techniques for BLDC motors in EV applications, focusing on the HCC, FOC, hybrid FOC+HCC, and FOC+HCC+SMC approaches. The controllers are modeled and simulated in MATLAB/Simulink, and their performance is evaluated based on speed tracking, torque ripple, transient response, and efficiency. The remainder of this paper is organized as follows: Section II discusses the system description, and Section III describes the system design. Section IV describes the system setup and its functioning. Section V describes the motor drive system. Section VI discusses different sensor-based control methods for BLDC motors; Section VII presents the simulation and results. Conclusions are given in Section VIII.

2 System description

Figure 3 illustrates the proposed system's comprehensive framework for a PV-based electric vehicle powertrain, including essential components such as the PV array, dual-active bridge (DAB) converter, battery storage, and motor control logic. The process

begins with energy harvesting from the PV array, with the help of a novel, robust MPPT hybrid quasi-slime mold and adaptive fish swarm optimization (HQSM-AFSM) technique, which is then regulated and converted into usable DC power through a DC–DC converter. An intensified Marine Predator Algorithm (MPA) for optimizing the design of solar-powered BLDC motors in EV systems, enhancing performance, efficiency, and convergence speed under dynamic operating conditions [22]. A detailed design and analysis of high-gain DC–DC chopper topologies for PV-fed BLDC motor drive systems, enabling superior voltage amplification, efficient energy conversion, and reliable operation in solar-based electric vehicle applications [23]. The grid will provide AC power, which is converted into usable DC power through a novel feed-forward decoupling control method. A DC/DC converter further optimizes the power flow between the grid and the PV system, as shown in Fig. 4. The filtered output charges the EV battery, which serves as the vehicle's primary energy source. The motor control section uses advanced controllers and a MOSFET-based drive circuit to manage the BLDC motor, ensuring efficient propulsion with minimal energy loss. The system controller applies advanced motor control methods—(FOC + SMC + HCC)—for precise speed and torque of the BLDC motor. Hall sensor feedback ensures accurate motor performance and dynamic

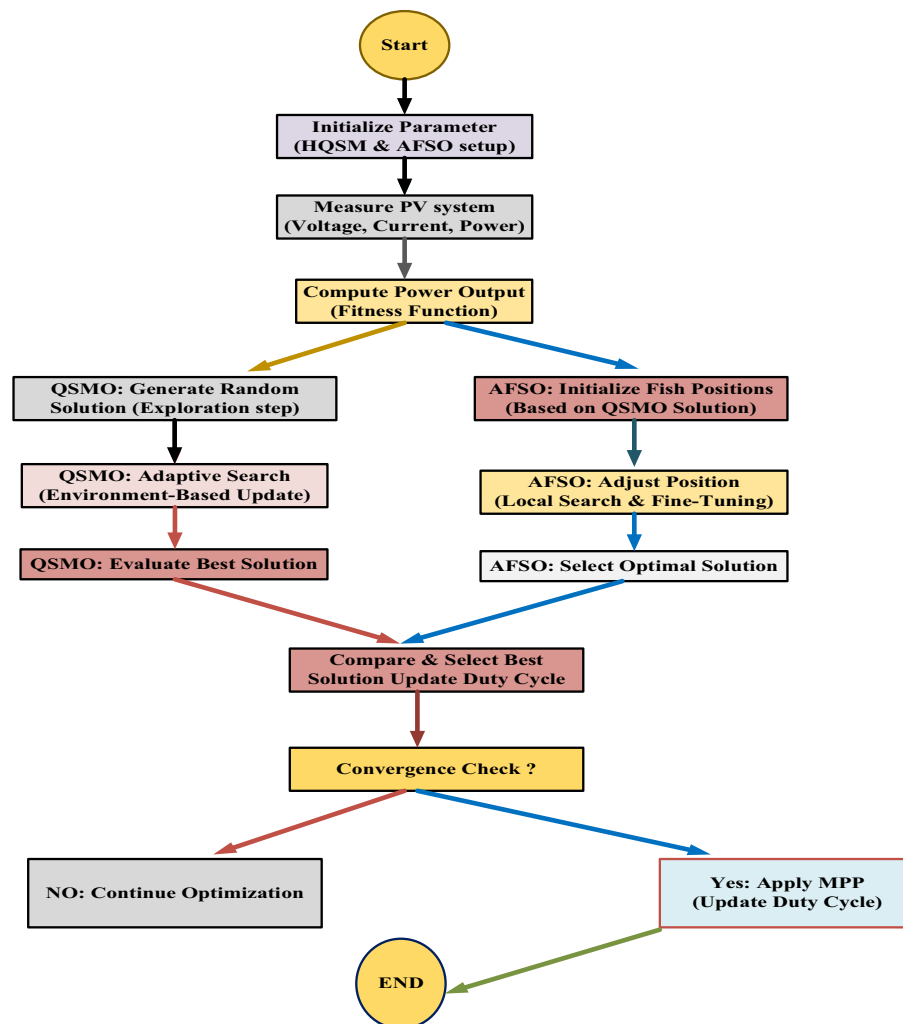


Fig. 4 Flow- chart of proposed Hybrid Q-SMA & AFSO MPPT Algorithm

response. These techniques enable efficient power management and smooth motor operation for reliable EV performance.

3 System design

The system configuration, illustrated in Figs. 3, 7, comprises several key components: the photovoltaic (PV) array, boost converter, three-phase grid, dual active bridge converter (DAB), three-phase voltage source inverter (VSI), and brushless DC (BLDC) motor. Each component is meticulously selected and designed to ensure optimal system performance. The PV array and the boost converter maximize energy utilization, while the grid and solar PV work together to maintain a stable DC link. The three-phase VSI efficiently powers the BLDC motor, enabling smooth control over speed and torque. This section outlines the detailed design considerations for each component.

3.1 Design of photovoltaic (PV) array

An 8-kWp photovoltaic (PV) array is designed to power a 5-kW brushless DC (BLDC) motor for EV application, considering the power losses in both the motor and the associated converters. The system parameters were established under standard test conditions, which included an irradiance of 1000 W/m², and a temperature of 25 °C. To ensure the required capacity, we selected a PV module with a maximum power point (MPP) voltage of 30.7 V and an MPP current of 8.15 A (detailed specifications are provided in the appendices). The MPP voltage of the PV array was set to align with the DC bus voltage of the BLDC motor, which corresponds to the DC voltage rating of the voltage source inverter (VSI). For this system, the MPP voltage was set to 430 V, and the remaining parameters were calculated accordingly.

The current level is at the maximum power point.

$$I_{mpp} = i_{pv} = \frac{p_{pv}}{v_{pv}} = \frac{8000}{430} = 18.60 \text{ A} \quad (1)$$

where, $P_{mpp} = p_{pv} = 8000$ Watt is the output power of a photovoltaic (PV) array at its MPP.

Series modules (N_s) are

$$N_S = \frac{V_{mpp}}{V_m} = \frac{430}{30.7} = 14 \quad (2)$$

Parallel modules (N_p) are

$$N_P = \frac{I_{mpp}}{I_m} = \frac{18.60}{8.15} \cong 3 \quad (3)$$

where, I_m and V_m are the MPP-current and MPP-voltage of the module. So, from Eqs. (2) and (3), 14 modules are connected in series and 3 modules are connected in parallel strings to design the photovoltaic (PV) array.

3.2 Design of DC–DC (boost) converter

The design of the boost converter involves determining the input inductor L , which is chosen to ensure that the converter operates in continuous conduction mode (CCM) under all weather conditions. The duty cycle D_1 is calculated as outlined in [24].

$$D_1 = \frac{V_{dc} - V_{pv}}{V_{dc}} = \frac{600 - 430}{600} = 0.283 \quad (4)$$

where, $V_{dc} = 600$ V is the DC link voltage across the DAB converter.

The value of the inductor (L) is calculated as

$$L = \frac{D_1 V_{pv}}{f_{sw} \Delta I_L} = \frac{0.283 * 430}{10000 * (18.60 * 0.2)} = 3.3mH \quad (5)$$

Here, f_{sw} represents the switching frequency of the boost converter, while ΔI_L the current ripple through the inductor L, with I_L being equal to the current at the maximum power point I_{mpp} .

3.3 Design of DC-link common capacitor

The DC-link capacitor C is shared among the boost converter, grid, and dual active bridge converter (DAB). It is designed to filter the second harmonic component of the 3-phase grid voltage that appears on the DC bus of the DAB converter. The value of the capacitor C is calculated as outlined in [25].

$$C = \frac{I_{dc}}{2w_L \Delta V_{dc}} = \frac{8000/600}{2 * (2\pi * 50) * (600 * 0.008)} = 100\mu F \quad (6)$$

where, I_{dc} represents the average current that flows through the DC bus, w_L signifies the line frequency measured in radians per second, and δV_{dc} refers to the voltage ripple on the DC bus.

3.4 Design of a three-phase voltage source inverter (VSI)

A three-phase VSI is used to feed the BLDC motor. Its design consists of an estimation of voltage, current, and volt-ampere (VA) ratings. As the DC bus voltage is 600 V, the battery voltage is 72 V, and the battery current is 70 Amp, the required voltage rating of an insulated gate bipolar transistor (IGBT) switch is calculated as [26].

$$V_{VSI} = V_{battery} \times 1.4 = 72 \times 1.4 = 100.8V \quad (7)$$

To account for switching transients, a voltage safety factor of 1.4 has been selected. Similarly, the current rating for the IGBT switch has been determined [26].

$$I_{VSI} = I_{battery} \times 1.3 = 70 \times 1.3 = 91A \quad (8)$$

The final step in determining the VA rating of the VSI is its evaluation

$$VA_{VSI} = V_{VSI} \times I_{VSI} \cong 9kVA \quad (9)$$

3.5 Design of the Inductor for (DAB) converter

In this section, we design the inductor (L) for the DAB converter. As illustrated below, the required inductance value for maximum power transfer to the output is determined using Eq. (10) [27].

$$P_{1\max} = \frac{0.129DV_{DC}V_b}{fL} \quad (10)$$

$$L = \frac{0.129DV_{DC}V_b}{fP_{1\max}} \quad (11)$$

$$L = \frac{0.129 \times 0.12 \times 600 \times 72}{100000 \times 5000} = 13\mu H \quad (12)$$

Therefore, an inductance of 13 μ H, which includes the leakage inductance of the HFT, is required for a power rating of 5 KW.

4 System setup and its functioning

This section discusses the DAB converter's design, steady-state performance, and solar MPPT technique. The converter facilitates efficient power transfer between the solar and grid to the battery, through a novel MPPT technique and feedforward decoupling control, enhancing grid stability.

4.1 Topology of DAB converter

The single-phase DAB converter-based battery charging system, as illustrated in Fig. 7, comprises two H-bridge circuits (H-Bridge-1 and H-Bridge-2), an inductor (L), and a high-frequency transformer (HFT). Each H-bridge consists of four self-commutating power semiconductor switches (IGBTs or MOSFETs). The inductor (L) represents the transformer's equivalent leakage inductance on the input side. If the leakage inductance of the HFT is insufficient, an additional inductor can be connected in series with (L) to achieve the necessary passive element value for efficient power transfer to the battery or load. This inductor (L) plays a crucial role in facilitating energy transfer within the converter. The DC input voltage and current are denoted as V_{in} and I_{in} , while the battery voltage and current are labeled as V_b and I_b . The primary and secondary voltages of the HFT are represented as V_{ab} and V_{cd} , respectively.

The DAB converter efficiently converts DC input (V_{in}) into an AC square waveform (V_{ab}) using H-Bridge-1, where switches (S1, S4) and (S2, S3) alternate during each half of the switching period. This AC voltage is transferred through a high-frequency transformer (HFT) for galvanic isolation and voltage scaling. On the secondary side, H-Bridge-2 converts the AC back to DC to charge the battery. The SPS modulation technique controls power transfer by coordinating both H-bridges, ensuring stable and efficient energy conversion for reliable battery charging.

4.2 Steady-state (SS) operation of DAB converter

Before conducting a steady-state analysis, several assumptions are made, including treating power semiconductor switches and passive components as ideal. Figure 8 illustrates the various waveforms of the DAB converter parameters alongside their corresponding switching signals. During a single switching cycle, four distinct operating modes can be identified based on the switching states of the different switches, as shown in Fig. 8. These modes are categorized according to the variation in inductor current, which is influenced by the phase shift between the two bridges. The total cycle duration is denoted as T_s , while half of this period is denoted as T_{sh} , where $T_{sh} = T_s/2$. Additionally, dT_{sh} represents the phase delay between the two H-bridges, where the parameter d represents the duty ratio, which serves as a control variable to regulate the phase shift between two successive bridges. The leakage inductance current (i_L) is

affected by the voltage applied across the inductor. When the voltage is positive, the current increases, while it decreases when the voltage turns negative. A detailed mathematical analysis of the DAB converter for each operating mode is provided in the following subsections.

4.2.1 Mode I ($0 \leq t \leq T_{sh}$)

In this mode, switches S1 and S4 of H-bridge-1 and S6 and S5 of H-bridge-2 are turned ON, while the others remain OFF. The inductor voltage ($V_{in} + nV_b$) is positive, leading to an increase in current. Equivalent circuits for $i_L < 0$ and $i_L > 0$ are shown in Fig. 8a, b. The corresponding voltage and current equations are provided below.

$$V_{in} = v_{ab}; -V_b = v_{cd}; v_L = nV_b + V_{in}; i_L = \frac{nV_b + V_{in}}{L}t - I_{L\min} \quad (13)$$

where, $I_{L\min}$ represents the initial current at the beginning of Mode 1. This mode continues until the time instant $t = dT_{sh}$, marking its completion.

4.2.2 Mode II ($dT_{sh} \leq t \leq T_{sh}$)

In this mode, H-bridge-1 operates with switches S1 and S4 turned ON, while in H-bridge-2, switches S5 and S8 are active, with all others OFF. Zero-voltage switching (ZVS) occurs during the dead time between switching transitions. The voltage and current equations for this mode are given below.

$$V_{in} = v_{ab}; -V_b = v_{cd}; v_L = nV_b + V_{in}; i_L = \frac{nV_b + V_{in}}{L}(t - dT_{sh}) - i_L(dT_{sh}) \quad (14)$$

During this period, V_{ab} remains positive and equal to V_{in} , while V_{cd} is positive and equals V_b . The inductor current direction is shown in Fig. 8, and mode II ends at $t = T_{sh}$.

4.2.3 Mode III ($T_{sh} \leq t \leq T_{sh} + dT_{sh}$)

In this mode, switches (S5, S8) continue conducting, while (S3, S2) are turned ON, and the rest (S1, S4, S6) remain OFF. The corresponding circuit configuration is illustrated in Fig. 8d, e. The voltage and current equations for this mode are provided below.

$$-V_{in} = v_{ab}; V_b = v_{cd}; v_L = -(nV_b + V_{in}); i_L = -\frac{nV_b + V_{in}}{L}(t - T_{sh}) - i_L(T_{sh}) \quad (15)$$

During this phase, the inductor current decreases since the slope of (i_L) is negative. The conduction path for positive and negative currents is illustrated in, respectively.

4.2.4 Mode IV ($T_{sh} + dT_{sh} \leq t \leq T_s$)

In Mode- IV, switches (S2, S3) remain ON, while (S6, S7) begin conducting, and (S5, S8, S1, S4) remain OFF. Both the primary voltage (V_{ab}) and secondary voltage (V_{cd}) are negative. The inductor current (i_L) continues to decrease, following the slope given in the corresponding equation.

$$\begin{aligned} -V_{in} &= v_{ab}; -V_b = v_{cd}; v_L = nV_b - V_{in} \\ i_L &= -\frac{V_{in} - nV_b}{L}(t - dT_{sh} - T_{sh}) - i_L(dT_{sh} + T_{sh}) \end{aligned} \quad (16)$$

4.3 Hybrid Q-SMA and AFSO MPPT algorithm (proposed*)

The Hybrid Q-SMA and AFSO MPPT Algorithm offer a superior solution for optimizing photovoltaic (PV) power extraction compared to traditional MPPT techniques like perturb and observe (P&O), incremental conductance (INC), and particle swarm optimization (PSO). While P&O and INC are simple and widely used, they suffer from slow tracking speeds, steady-state oscillations, and reduced accuracy under rapidly changing environmental conditions. PSO improves tracking efficiency but can struggle with convergence speed and computational complexity. An enhanced PSO algorithm implemented on a dual-core DSP to improve MPPT performance in partially shaded PV systems. Their approach enables real-time tracking of the global maximum power point, significantly increasing energy efficiency and reliability in EV battery charging under varying solar conditions [28]. In contrast, the proposed hybrid approach leverages Q-SMA for enhanced global exploration and AFSO for adaptive fine-tuning, ensuring fast, stable, and precise MPPT tracking. This combination minimizes power losses, enhances performance under partial shading, and achieves higher efficiency, faster convergence, and reduced fluctuations compared to conventional algorithms. Thus, the Hybrid Q-SMA and AFSO MPPT algorithm provides a more reliable and intelligent solution for modern solar energy systems. An efficient torque control method for BLDC motors powered by solar PV systems using the Salp Swarm Optimization (SSO) algorithm. Their approach optimizes controller parameters to enhance motor performance, reduce torque ripple, and improve energy efficiency. The integration of SSO ensures fast convergence and robust tracking under variable irradiance, benefiting electric vehicle drive systems [10]. An advanced variable step size Maximum Power Point Tracking (MPPT) algorithms using Grey Wolf Optimization (GWO) and Whale Optimization Algorithm (WOA) for photovoltaic systems. These algorithms enhance power output by improving tracking accuracy and convergence speed. The study demonstrates superior performance in dynamic conditions, effectively maximizing energy extraction and system efficiency under variable solar irradiance and temperature fluctuations [29].

The mathematical modeling of the Hybrid Q-SMA and AFSO MPPT algorithm combines quadratic approximation with swarm-based optimization techniques. The power output at each time step is updated using:

PV Power Model:

$$P = V * I$$

where P = Output power of the PV panel, V = Output voltage, I = Output current.

Q-SMA Position Update (Quantum-based Sine Cosine Algorithm):

$$X_{t+1} = X_t + \alpha \cdot \cos(\beta) + \gamma \cdot \sin(\delta)$$

$$V_{new} = V_{old} + k_1 * \left(\frac{\Delta P}{\Delta V} \right) + k_2 * \left(\frac{\Delta^2 P}{\Delta V^2} \right) \quad (17)$$

where: X_t is the current position, α , β , γ , δ are control parameters, k_1 , k_2 = Algorithm tuning parameters, $\Delta P / \Delta V$ and $\Delta^2 P / \Delta V^2$ = First and second derivatives of power w.r.t. voltage.

Adaptive fractional swarm optimization (AFSO):

$$X_{i+1} = X_i + c_1 r(P_{best} - X_i) + c_2 r_2(G_{best} - X_i) \quad (18)$$

where X_i is the current particle position, c_1 , c_2 are learning coefficients, r_1 and r_2 are random numbers between 0 and 1, P_{best} is the personal best position, G_{best} is the global best position.

Hybrid MPPT Update Rule: The hybrid algorithm refines the step size dynamically:

$$V_{new} = V_{old} + \lambda \cdot (Q - SMA_{update} + AFSSO_{update}) \quad (19)$$

where λ is an adaptive factor controlling convergence speed.

4.4 Solar photovoltaic (PV) system

Figure 5 illustrates the I–V and P–V characteristics of a single-string PV array with 14 modules, considering 5 different irradiance levels: (1000, 300, 600, 400, 200) W/m^2 . These characteristics highlight the impact of varying sunlight intensity on the electrical performance of the PV system.

4.5 EV battery

This study selected a lithium-ion battery for use in electric vehicles, specifically a Lithium Iron Phosphate (LiFePO_4) battery designed for two-wheeled electric vehicles. The battery management system (BMS) is vital for ensuring the safe and efficient operation of lithium iron phosphate (LiFePO_4) batteries, particularly in electric vehicles (EVs), where thermal stability and long cycle life are paramount. It actively monitors and controls key parameters such as voltage, current, and temperature to prevent overcharging, over-discharging, and thermal runaway, thereby safeguarding battery integrity. Advanced protection mechanisms like real-time cell voltage balancing, temperature regulation, and fault detection (e.g., short circuit or overcurrent) ensure uniform performance across all cells in the pack. By managing the battery's state of charge (SOC) and state of health (SOH), the BMS optimizes energy usage and prolongs battery lifespan. These integrated

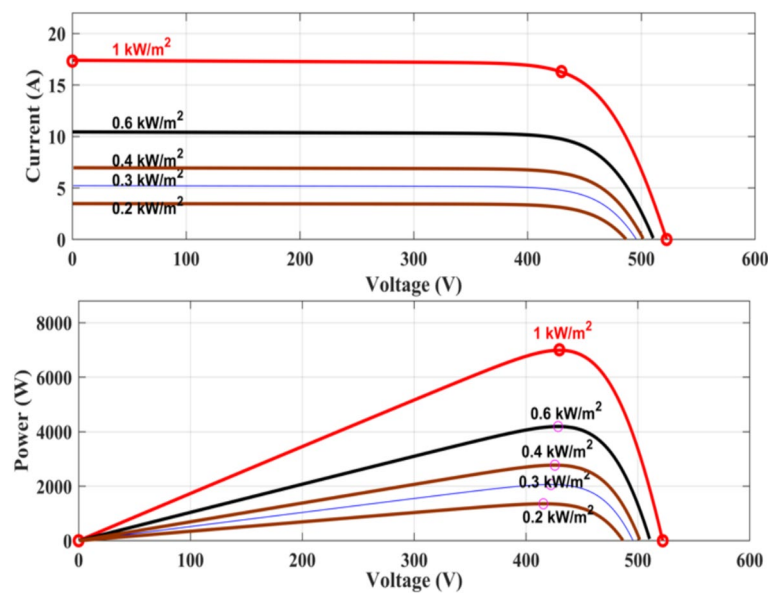


Fig. 5 The ideal characteristics of a PV string include: **a** I–V curve and **b** P–V curve across three different irradiance levels at (1000, 600, 400, 300, 200) W/m^2

Table 1 Specification of EV battery

Battery name	Meshi lithium battery
Model	72V20S
Model and capacity	10P 26Ah
Charging voltage (V)	80
Charging current (A)	62
Dimension	225*220*185 mm
SOC (%)	50
Battery response time (sec)	30

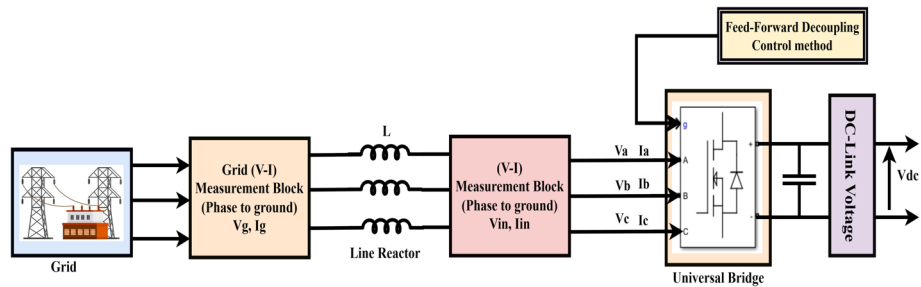


Fig. 6 Diagram of grid-side control method

safety functions collectively enhance the reliability, durability, and overall safety of LiFePO₄ batteries in high-demand EV environments (Table 1).

4.6 Feed-forward decoupling control

Figure 6 illustrates the hybrid renewable energy system that combines solar photovoltaic (PV) power and grid electricity. Maintaining a stable 600 V DC link voltage is critical for reliable operation and efficient power transfer. The feed-forward decoupling control strategy is designed to smartly manage power between a solar PV system and the grid. It gives first priority to solar power, especially when sunlight is strong, and only uses grid power when solar energy is not enough to meet the load. The DC link serves as an energy buffer, facilitating a seamless power flow between the solar PV system, the grid, and the load. This control ensures a smooth and stable switch between the two sources without affecting the motor's performance. The system continuously checks how much power is needed and how much is available from solar, adjusting the balance in real-time. In light load conditions, solar alone can handle the demand, while in heavy load situations, the grid helps meet the extra need. During medium loads, both sources work together efficiently to balance supply. The system provides a total power output of 5 kW, with contributions from both the solar PV system and the grid. The PV system operates under varying irradiance levels, which can range from 200 to 1000 W/m² and is managed by a boost DC–DC converter. When solar power generation is insufficient, the grid compensates for the shortfall. To ensure fast and stable voltage regulation, a feedforward decoupling control strategy is applied to the grid-side universal bridge, enabling bidirectional power flow. This approach maintains a steady 600 V DC link voltage, even in the face of fluctuations in solar generation and load demand, thus enhancing overall system stability and efficiency. This method improves energy use, reduces grid dependence, and keeps the system stable.

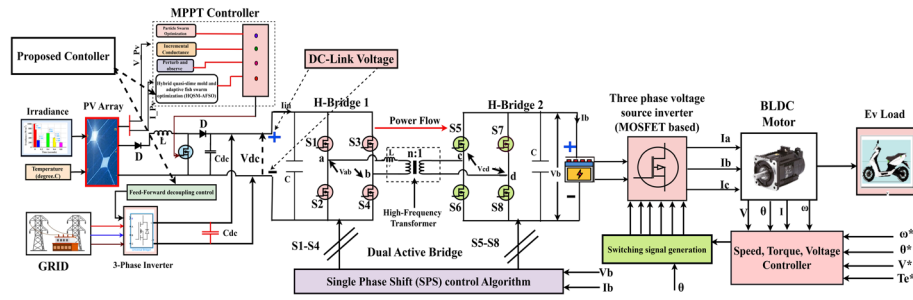


Fig. 7 An overview of solar-grid EV drive

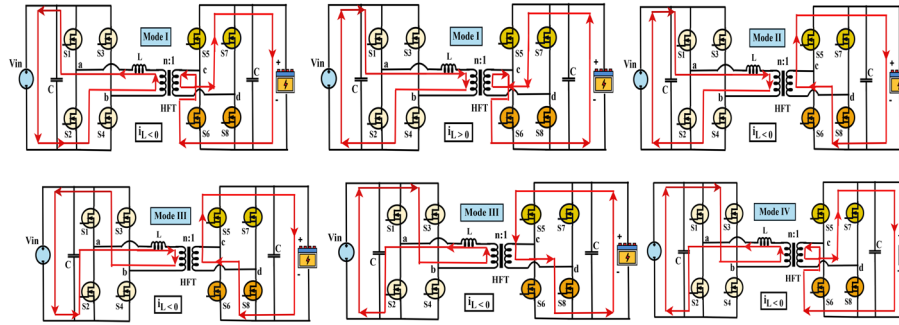


Fig. 8 The equivalent circuit for each operating mode illustrates the current flow path through the system

Where V_a , V_b , and V_c represent the 3-phase grid voltage (v_s) and i_b , i_a , and i_c , represent the 3-phase current, respectively. L is a line inductor or grid inductance, and V_{dc} is output voltage (v_t), as shown in Fig. 6.

Transforming from the abc reference frame to the dq synchronous rotating frame simplifies AC system analysis by converting three-phase time-varying signals into steady-state DC quantities [30] (Figs. 7, 8).

$$\begin{bmatrix} \frac{di_d}{dt} \\ \frac{di_q}{dt} \end{bmatrix} = \frac{1}{L} \begin{pmatrix} -R & \omega L \\ -\omega L & -R \end{pmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} - \frac{1}{L} \begin{bmatrix} v_{sd} \\ v_{sq} \end{bmatrix} + \frac{1}{L} \begin{bmatrix} v_{td} \\ v_{tq} \end{bmatrix} \quad (20)$$

where v_{sq} and v_{sd} represent the q -axis and d -axis components of the 3-phase grid voltage. i_q and i_d represent the q -axis and d -axis components of the currents respectively. v_{tq} and v_{td} represent the q -axis and d -axis components of the output dc-link voltage.

Equation (20) can be also rewritten as:

$$\begin{aligned} v_{td} &= L \frac{di_d}{dt} + Ri_d - L\omega_0 i_q + v_{sd} \\ v_{tq} &= Ri_q - L\omega_0 i_d + L \frac{di_q}{dt} + v_{sq} \end{aligned} \quad (21)$$

In the dq mathematical model given in Eq. (21), the presence of the $L\omega_0$ term causes coupling between the dynamics of i_d and i_q , making the control design more challenging. The voltage-sourced converter (VSC) can be represented using the following model.

$$\begin{aligned} v_{tq} &= m_q(t) \frac{V_{DC}}{2} \\ v_{td} &= m_d(t) \frac{V_{DC}}{2} \end{aligned} \quad (22)$$

The decoupling of these dynamics can be accomplished by defining $m_q(t)$ and $m_d(t)$ as follows [30].

$$\begin{aligned} m_d(t) &= \frac{2}{V_{DC}} (u_d - \omega L i_q + v_{sd}) \\ m_q(t) &= \frac{2}{V_{DC}} (u_q - \omega L i_d + v_{sq}) \end{aligned} \quad (23)$$

where, u_q and u_d are the two control inputs. Put m_q and m_d value in Eqs. (21 and 22), then it concludes that: -

$$\begin{aligned} L \frac{di_d}{dt} &= -R i_d + u_d \\ L \frac{di_q}{dt} &= -R i_q + u_q \end{aligned} \quad (24)$$

As a result, the control equations can be expressed as follows;

$$\begin{aligned} v_{tq} &= (i_q^* - i_q) \left(k_p + \frac{k_i}{s} \right) + \omega L i_d + v_{sq} \\ v_{td} &= (i_d^* - i_d) \left(k_p + \frac{k_i}{s} \right) - \omega L i_q + v_{sd} \end{aligned} \quad (25)$$

4.7 Power decoupling control

Based on the instantaneous power theory [31], in the dq synchronous rotating reference frame, the following expression can be derived (Table 2).

$$\begin{aligned} P &= \frac{3}{2} (v_{sd} i_d + v_{sq} i_q) \\ Q &= \frac{3}{2} (v_{sq} i_d - v_{sd} i_q) \end{aligned} \quad (26)$$

P is active, and Q is reactive power (Fig. 9).

Table 2 Comparison of proposed MPPT with existing MPPT algorithm

MPPT algorithm	Tracking speed	Steady-state oscillation	Convergence time	Simulation results	Efficiency	References
Perturb and Observe (P&O)	Slow	Low	Long	Slower convergence under rapidly changing conditions	85–95%	[28, 32]
Incremental Conductance (INC)	Moderate	Moderate	Moderate	Works well under steady-state conditions but struggles with rapid fluctuations	95–98%	[28, 33]
Particle Swarm Optimization (PSO)	Fast	High	Short	Faster convergence but may get trapped in local optimal	85–95%	[26, 28]
HQSM-AFSO (Proposed)*	Very Fast	Very Low	Very Short	Improved tracking speed and efficiency under varying load conditions	> 99%	[10, 29]

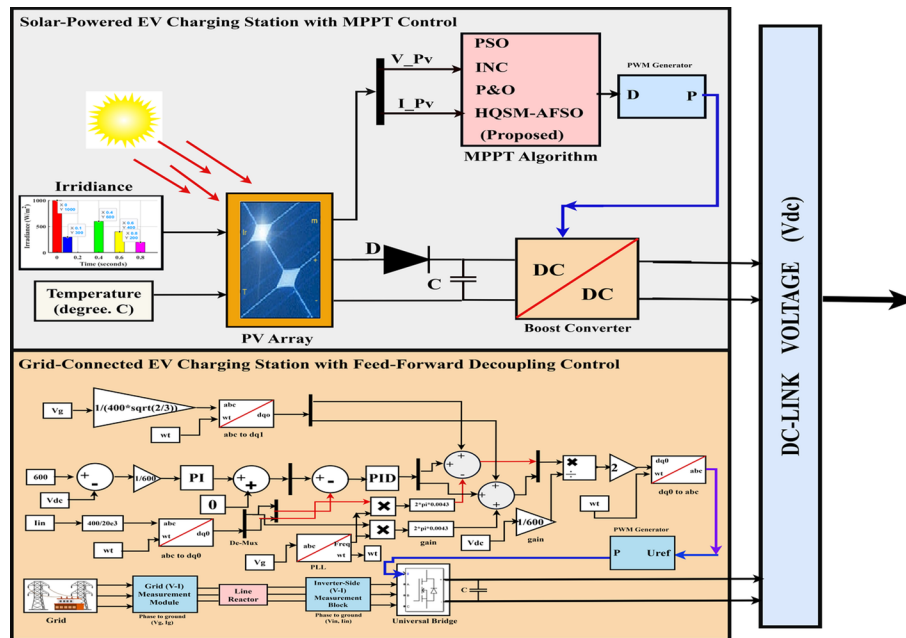


Fig. 9 The equivalent circuit for each operating mode illustrates the current flow path through the system

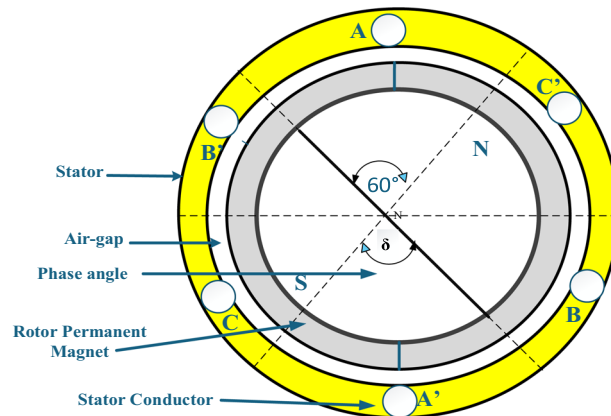


Fig. 10 Cross-section of an ideal Brushless DC Motor

5 Motor drive system

5.1 BLDC motor

A three-phase permanent magnet (PM) motor operates similarly to an induction motor, with a three-phase stator winding designed to produce an air-gap flux distribution that can be either sinusoidal or trapezoidal. (PMSM and BLDC), both motors come under the permanent magnet motor technology, but both motors will differ based on the interaction of back EMF with armature current. In the case of (PMSM), the interaction of sinusoidal back EMF with sinusoidal current, whereas interaction of trapezoidal back EMF with the quasi-square waveform of armature current should take place in BLDC motor. Figure 10 illustrates a basic two-pole motor design with a 180-degree magnetic pole arc. The stator winding consists of three phases with each phase occupying two slots. In this configuration the electrical phase angle (δ) between two adjacent coils is 60° .

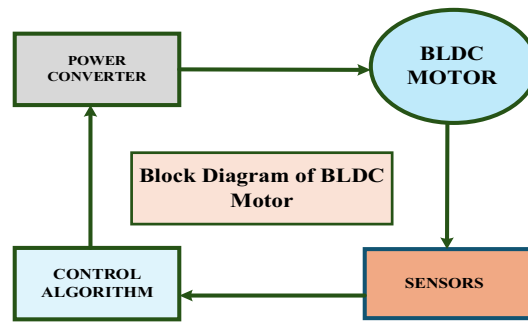


Fig. 11 Block illustration of BLDCM drive

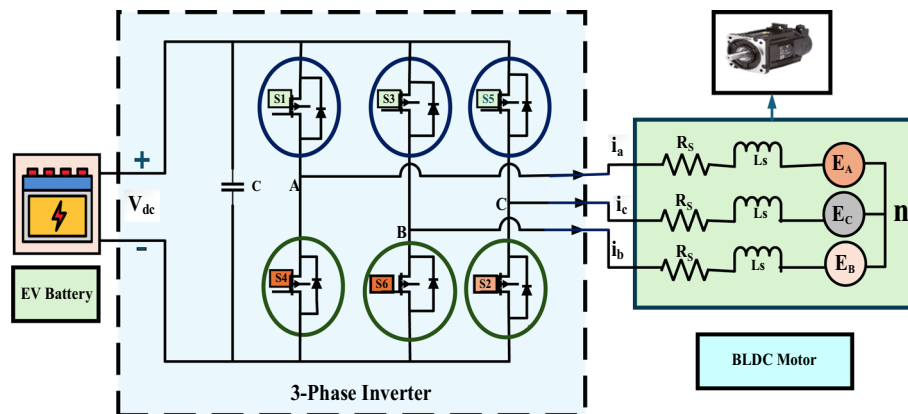


Fig. 12 Simplified model illustration of BLDCM drive

5.2 Block diagram of BLDC motor

The fundamental structure of the BLDCM drive system is illustrated in Fig. 11. This system comprises four key components; a power converter that delivers power to the motor, the brushless direct current (BLDC) motor itself, detectors/sensors such as hall sensors, encoders, and speed sensors that provide input data to the controller, and a control strategy that produces control signal for the power converter to control the motor's speed and power output towards meet the desired performance.

5.3 Modelling of BLDC motor

The fundamental equation for voltage and torque in BLDC motors are comparable to those used in brushed DC motors. Figure 12 depicts a simplified model illustration of a BLDCM drive powered by a battery through a 3-phase inverter composed of 6 MOSFET switches. The rotor, equipped with a permanent magnet, causes the flux linkage to be influenced by the magnet, making magnetic flux saturation a key characteristic of BLDC motors. The assumptions are as follows:

1. The motor operates within its specified current limits to prevent saturation.
2. The 3-phases are evenly distributed.

The BLDC motor has a star-connected stator winding.

The voltage equations are expressed as follows:

$$V_a = e_a + Ri_a + L_a \frac{di_a}{dt} \quad (27)$$

$$V_b = e_b + Ri_b + L_b \frac{di_b}{dt} \quad (28)$$

$$V_c = e_c + Ri_c + L_c \frac{di_c}{dt} \quad (29)$$

where, $L_a = L_c = L_b = L = L_S - L_M$ [Henry].

$$R_a = R_b = R_c = R$$

L_S , L_M , and R are the stator self-inductance, mutual-inductance, and armature resistance in ohms. Also, i , V , e , $\emptyset$, Θ , and ω are phase input motor current, the terminal-phase voltage of the motor in (V) volts, and the back-EMF of the motor in volts, the magnetic flux generated by the permanent magnet, rotor position angle and angular speed, respectively.

The representation of back emf can be expressed as follows:

$$e_a(t) = \omega(t) * \emptyset(\theta) * K_E \quad (30)$$

$$e_b(t) = \omega(t) * \emptyset\left(\theta - \frac{2\pi}{3}\right) * K_E \quad (31)$$

$$e_c(t) = \omega(t) * \emptyset\left(\theta + \frac{2\pi}{3}\right) * K_E \quad (32)$$

As shown below, it can be represented in a compact matrix form.

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} L & 0 & 0 \\ 0 & L & 0 \\ 0 & 0 & L \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix}$$

where i_b , i_c , and i_a are input phase currents, V_b , V_a , and V_c represent terminal-phase voltages. M represents mutual inductance between 2-phases. As $i_b + i_c + i_a = 0$, the inductance matrix is modified as explained in (8).

$$\begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \begin{bmatrix} L & 0 & 0 \\ 0 & L & 0 \\ 0 & 0 & L \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (33)$$

The produced torque is expressed as follows:

$$\tau_e(t) = \frac{(e_a i_a + e_b i_b + e_c i_c)}{\omega} \quad (34)$$

$$\tau_a(t) = i_a(t) * \emptyset(\theta) * K_T \quad (35)$$

$$\tau_a(t) = i_b(t) * \emptyset\left(\theta - \frac{2\pi}{3}\right) * K_T \quad (36)$$

$$\tau_a(t) = i_b(t) * \emptyset\left(\theta + \frac{2\pi}{3}\right) * K_T \quad (37)$$

where K_T denotes the Torque constant.

Assuming no phase difference exists between the current and the back-emf, the electromagnetic torque can be expressed as:

$$T_e = \frac{2EI}{\omega} \quad (38)$$

where I and E are DC-link current amplitude and back-EMF voltage peak amplitude.

Additionally, the electromagnetic torque can be represented as:

$$T_e = T_L + J \frac{d\omega}{dt} + B\omega \quad (39)$$

where J represents rotor inertia [kgm^2], T_L is load torque (N-m), and B represents the damping constant.

6 Different sensor-based control methods for BLDC motors

This section will explain the different closed-loop control techniques studied for BLDC motor control. The parameters controlled in this work's case of BLDC motor drive are rotational speed and electromagnetic torque. In the close loop, control electromagnetic torque is controlled through stator phase currents because electromagnetic torque is proportional to stator current.

6.1 HCC control

Figure 13 illustrates a block diagram of the (PI+HCC) controller. The stator current is monitored and evaluated to the reference current to generate a deviation/error signal. This signal is then directed into a comparator with a defined hysteresis range. The switching of the MOSFETs is determined by comparing the stator current and the reference current. A typical sampling rate for HCC is around 10–20 kHz, depending on the motor speed and load conditions.

A quasi-square wave pattern is achieved by controlling the desired current. The computational load of HCC is relatively low compared to more advanced control techniques. It mainly involves comparing the actual current with the reference value and switching the voltage accordingly, making it computationally efficient. The current error is

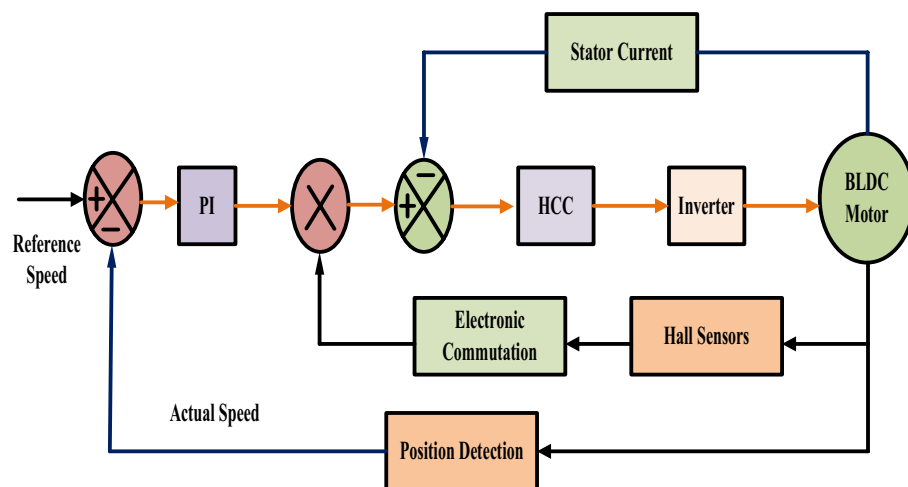


Fig. 13 BLDCM drive operation with (PI+HCC) controller

$\Delta I_a = I_a^* - I$, where 'L' and 'H' illustrate the lower and upper limits of the specified hysteresis bands. The switching sequence for the MOSFETs is found in Table 3.

6.2 Field oriented control (FOC)

FOC demonstrates that synchronous motors, including BLDC motors, can be controlled similarly to separately excited DC motors. The flux and torque-producing components in DC motors are maintained at a 90° angle, enabling independent control. FOC requires a moderate to high sampling frequency to maintain precise motor control, typically in the range of 5–10 kHz. FOC addresses this by independently regulating torque and flux through mathematical transformation. FOC is computationally more demanding than HCC because it involves coordinate transformations (Clark and Park transforms), real-time motor model updates, and vector calculations. Field Oriented Control (FOC) for a traction drive system powered by a photovoltaic (PV) and battery energy storage system (BESS), designed specifically for electric vehicle (EV) applications. Their approach enables accurate torque control, improved dynamic response, and enhanced overall drive efficiency. The integration of PV-BESS ensures reliable energy supply under variable load and environmental conditions [34].

The Clarke transform translates 3-phase variables into a 2-phase orthogonal system, simplifying analysis. The Park transform then circulates this 2-phase system into a reference framework aligned with the rotor's magnetic field.

Clark transformation:

$$I_\alpha = I_a I_\beta = \frac{1}{\sqrt{3}} I_a + \frac{2}{\sqrt{3}} I_b \quad (40)$$

Park transformation:

$$I_d = I_\alpha \cos\theta + I_\beta \sin\theta \quad I_q = I_\beta \cos\theta - I_\alpha \sin\theta \quad (41)$$

Inverse Park transformation

$$V_\alpha = V_d \cos\theta - V_q \sin\theta \quad V_\beta = V_d \sin\theta + V_q \cos\theta \quad (42)$$

Inverse Clark transformation:

$$V_\alpha = V_a V_b = -\frac{1}{2} V_\alpha + \frac{\sqrt{3}}{2} V_\beta \quad V_c = -V_\alpha \frac{1}{2} - V_\beta \frac{\sqrt{3}}{2} \quad (43)$$

FOC in AC synchronous motors independently regulates torque and flux by managing the direct (I_d) and quadrature (I_q) axis currents. I_d controls magnetic flux, while I_q generates torque. This decoupling enables fast response to speed changes and full torque at

Table 3 Switching sequence

Condition	ON	OFF
$\Delta I_a > H$	S1	S2
$\Delta I_a < L$	S2	S1
$\Delta I_b < L$	S4	S3
$\Delta I_b > H$	S3	S4
$\Delta I_c > H$	S5	S6
$\Delta I_c < L$	S6	S5

zero speed. By minimizing I_d , FOC maximizes efficiency as all current contributes to torque production via I_q (Fig. 14).

6.3 Field-oriented controller using hysteresis current controller (FOC + HCC)

In a modified FOC system, three-phase reference currents are generated by setting $i_{d(\text{ref})}$ to zero with a PI controller determining $i_{q(\text{ref})}$ from the speed error. The d-q reference frame currents are transformed into a stationary two-axis frame using the inverse Park transformation, followed by the inverse Clarke transformation to obtain three-phase “abc” currents. The reference and actual currents are compared to produce error signals fed to the hysteresis controller that controls the inverter switches. This controller effectively minimizes torque ripples by regulating the stator current within specified limits, ensuring smooth electromagnetic torque control. The sampling frequency for FOC + HCC is generally similar to FOC, typically around 5–10 kHz. The integration of HCC with FOC requires careful synchronization between the current control and field orientation algorithms. Figure 15 depicted the block diagram of the modified FOC using the HCC controller. The FOC + HCC method increases the computational load compared to FOC alone due to the additional hysteresis current control loop. However, the overall computational requirements are still manageable for modern control platforms.

The switching logic implemented in the method is found in Table 4.

6.4 Field field-oriented controller using hysteresis current controller and sliding mode controller (FOC + HCC + SMC) *

Figure 16 shows the modified FOC technique that uses SMC instead of PI controller to generate $i_{q(\text{ref})}$. SMC is a type of control system design that aims to achieve robust and efficient control of a system in engineering applications. The control law is designed to take the system to a hyperplane in the state space called “sliding mode,” which, once reached, forces the system to remain on it, resulting in robust and efficient control. This control is beneficial when the system faces uncertainties or disturbances, as it is designed to be strong to those factors. Integral sliding control (ISM) includes an additional

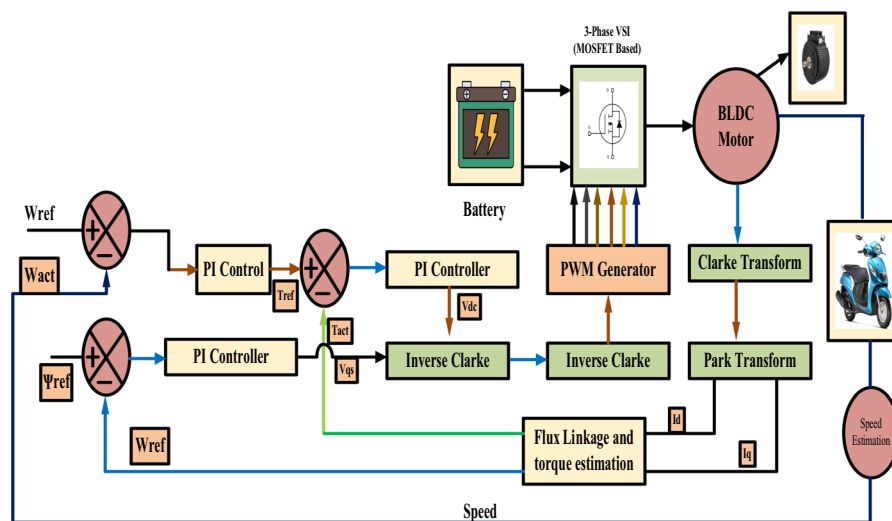


Fig. 14 Block diagram of conventional FOC

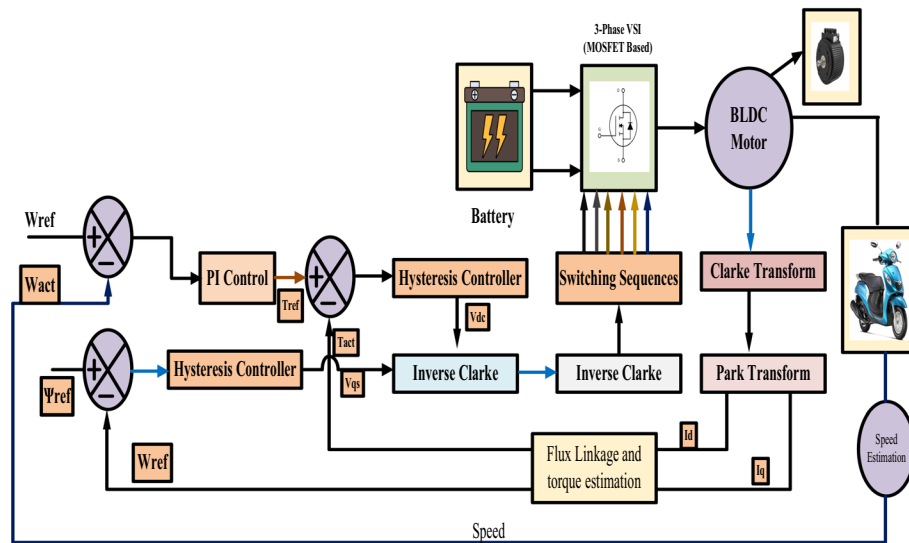


Fig. 15 Block diagram of modified field-oriented control using HCC

Table 4 Switching logic of different phases

Phase	Condition	S1	S4
A	$i_a < (i_a(\text{ref.}) - h)$	ON	OFF
	$i_a > (i_a(\text{ref.}) - h)$	OFF	ON
	Condition	S3	S6
B	$i_b < (i_b(\text{ref.}) - h)$	ON	OFF
	$i_b > (i_b(\text{ref.}) - h)$	OFF	ON
	Condition	S5	S2
C	$i_c < (i_c(\text{ref.}) - h)$	ON	OFF
	$i_c > (i_c(\text{ref.}) - h)$	OFF	ON

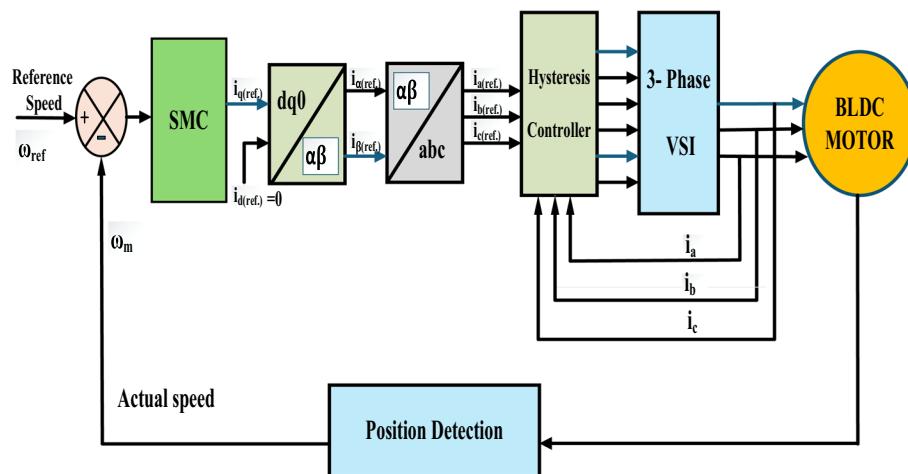


Fig. 16 Proposed block diagram of modified FOC using SMC and HCC

integral term in control laws and reduces the chattering effect without affecting high-frequency disturbances and steady-state errors should be zero. The FOC + HCC + SMC method requires a higher sampling frequency to ensure precise control and to handle the additional complexities of sliding-mode control. A sampling rate of 10–20 kHz is

typically required for optimal performance. An Adaptive Sliding Mode Control (ASMC) approach integrated with Maximum Power Point Tracking (MPPT) for enhancing the performance of a boost converter in photovoltaic (PV) systems. Their study focused on improving tracking accuracy and dynamic response under varying environmental conditions, such as fluctuating irradiance and temperature. The proposed ASMC-based MPPT method outperformed traditional controllers by offering faster convergence to the maximum power point and better robustness against system disturbances [35].

ω_m is taken as rotor speed and ω_{ref} as reference speed. The speed error (er) can be defined as $er = \omega_m - \omega_{ref}$. SMC takes two inputs, $\times 1$ and $\times 2$. One is an error, and the other is the rate of change of error. The computational load for FOC+HCC+SMC is the highest among the techniques evaluated. This method combines field-oriented control, hysteresis current control, and sliding-mode control, each of which requires real-time computations. The complexity of the sliding-mode controller, particularly in terms of calculating the switching surfaces and ensuring system stability, increases the computational burden. Therefore, a high-performance processor is necessary for real-time implementation (Fig. 17).

If $x_1 = e$ and $x_2 = \dot{x}_1 = \dot{e}$, then $s = ax_1 + x_2$ if $s = 0$ that is sliding surface equal to zero, then $x_1(t) = x_1(t) e^{-at}$. For $a > 0$, $x_1(t)$ will converge to zero, and the mean error will converge to zero. $\times 2$ is a scaled function of $\times 1$.

7 Result and discussion

The results below present the DC-link voltage, battery parameters, grid parameters, and a comparative analysis of different MPPT algorithms and four distinct controllers for controlling the torque and speed ripples of the BLDC motor. Parameters like speed, back-emf, stator current, and torque ripples of the motor are analyzed for four distinct controllers while maintaining all other motor specifications unchanged. The motor specifications are presented in Table 5 and used for this study. The results were generated using the MATLAB/SIMULINK environment.

The given Fig. 18 compares different MPPT algorithms PSO, INC, P&O, and the proposed HQSM-AFSO in tracking the maximum power point (MPP) of a photovoltaic (PV) system. Initially (0–0.1 s), all algorithms experience transient oscillations, but HQSM-AFSO (orange line) quickly stabilizes, showing a faster response than the others. PSO (blue line) reacts slower, while INC (black line) and P&O (brown line) have significant fluctuations. Midway (0.4–0.6 s), HQSM-AFSO maintains higher voltage with minimal oscillations, proving its efficiency during dynamic irradiance changes. By the final stage (0.8–1.0 s), HQSM-AFSO sustains better performance, while PSO stabilizes slowly, and INC & P&O suffer from poor tracking efficiency. With the shortest settling time

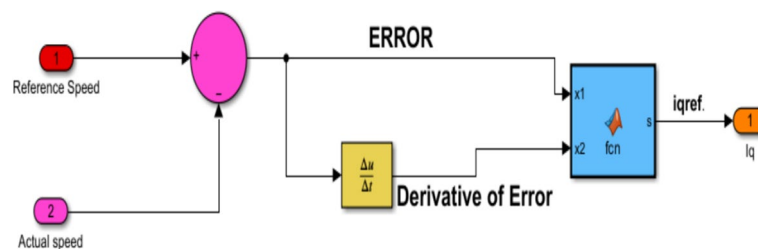


Fig. 17 MATLAB diagram of forming $I_{q(ref)}$ using SMC

Table 5 BLDC motor specification

Parameter	Value
Input DC voltage	72 V
Stator resistance	0.94 Ω
Stator inductance	0.0085H
Rated torque	12N-m
Rated power	5000 W
Rated current	70 A
Poles-pairs	2
Moment of inertia	0.8*10 ⁻³ kg-m ²
Rated speed	3000 rpm

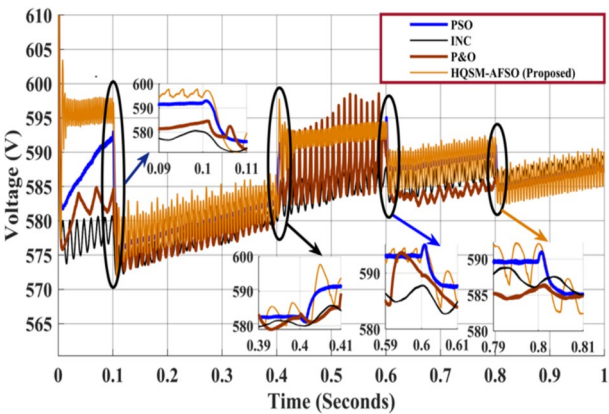


Fig. 18 Voltage tracking performance of different MPPT algorithms under dynamic condition

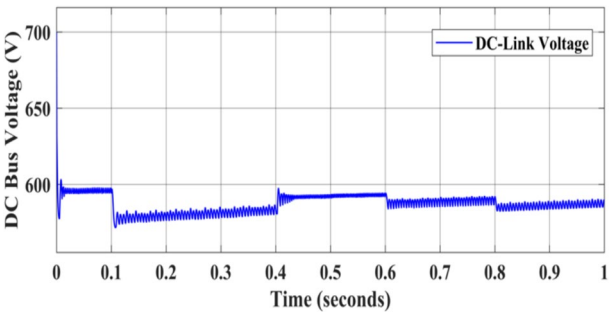


Fig. 19 DC-link voltage profile of a solar and grid-connected system

(~0.1–0.15 s) and superior adaptability, HQSM-AFSO outperforms traditional MPPT methods, making it ideal for real-time solar applications.

Figure 19 illustrates the DC-link voltage profile of a solar and grid-connected system, where a boost converter and a feedforward decoupling control technique maintain the voltage at a constant value of (~600V). The plot shows the system's ability to sustain a stable DC-link voltage under varying load conditions and power inputs from both the solar source and the grid. Initial transient oscillations are observed as the system stabilizes, followed by consistent voltage regulation throughout the simulation period.

Figure 20 illustrates the variations in PV voltage, current, and power over time, highlighting the system's response to changing conditions. Initially, the PV voltage rises sharply and then stabilizes at around 400–450 V, with transient fluctuations occurring

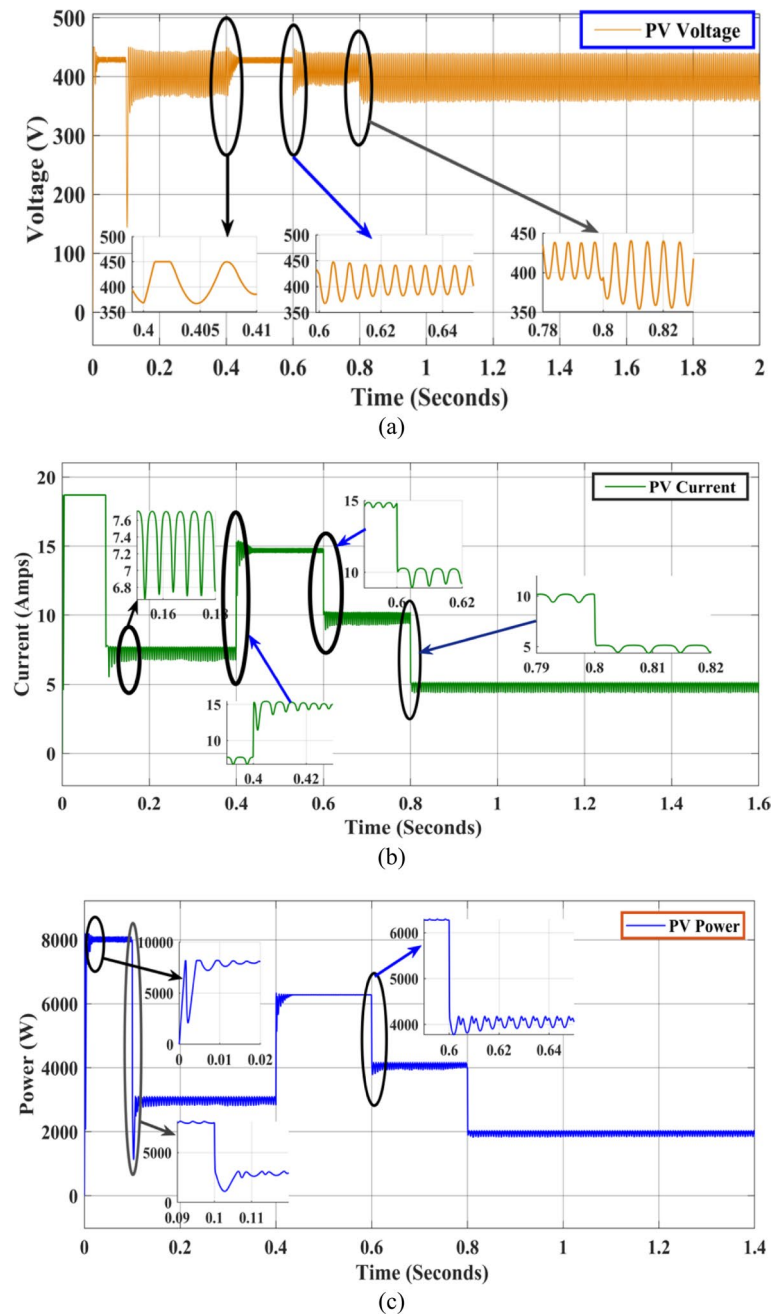


Fig. 20 **a** PV voltage, **b** PV current, **c** PV power curve across three different irradiance levels at (1000, 600, 400, 300, 200) W/m^2

at 0.4, 0.6, and 0.8 s due to load changes or MPPT adjustments. Similarly, the PV current experiences oscillations, showing sudden drops and recoveries at specific points, indicating dynamic power regulation. The PV power graph exhibits an initial low output, followed by a significant increase around 0.6 s, suggesting improved efficiency through MPPT optimization. The zoomed-in sections highlight critical voltage, current, and power instabilities, emphasizing system adaptability. All parameters stabilize after 1.2–1.4 s, ensuring a steady and reliable power supply. These variations indicate the system's ability to adjust to external factors like solar irradiation changes. The control mechanisms effectively regulate fluctuations and enhance overall system performance. The

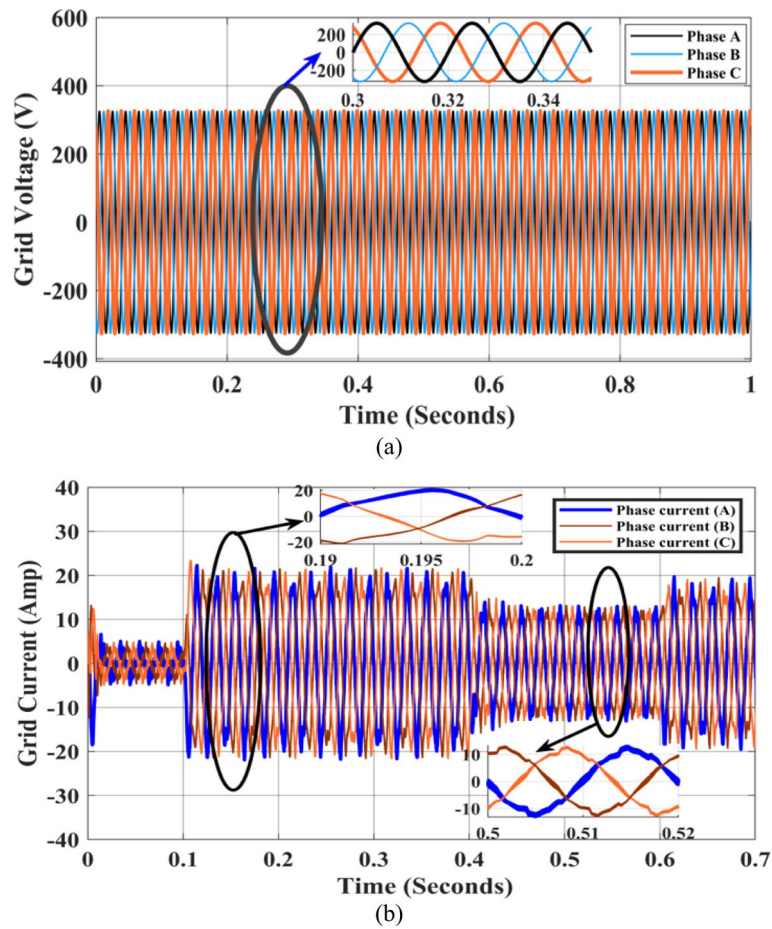


Fig. 21 a Grid voltage, b Grid current

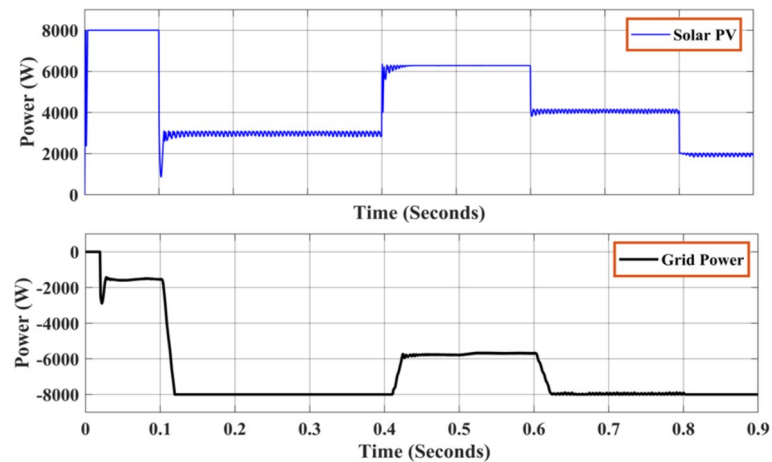


Fig. 22 Dynamic power sharing between solar PV and grid

MPPT algorithm plays a key role in maximizing energy extraction. Overall, the system demonstrates robust stability, adaptability, and efficiency in photovoltaic energy conversion (Fig. 21).

Figure 22 illustrates the power-sharing dynamics between solar PV power and grid power over time. The top subplot (blue curve) represents the solar PV power output,

while the bottom subplot (black curve) shows the grid power contribution. Initially, solar PV power rises to 8000 W but then drops to 2000 W before stabilizing, indicating MPPT-based optimization. At 0.4 s, PV power increases to 6000 W, showing improved energy harvesting. After 0.6 s, a slight drop suggests dynamic power management responding to load variations. Conversely, grid power initially drops to -8000 W, meaning the grid supplies power to balance fluctuations. Around 0.4 s, grid power reduces, indicating that PV power is taking a larger load share. By 0.6 s, grid power stabilizes near 0 W, implying reduced grid dependency. The figure highlights an efficient power management strategy, ensuring a stable power supply. This interaction demonstrates the effectiveness of hybrid renewable energy systems in improving efficiency and grid independence (Fig. 23).

The first one illustrates the controlled variable, especially the mechanical speed in Fig. 24. An elliptical signal was used as the reference speed to facilitate a thorough study and analysis of the BLDC motor's behavior, allowing observation of the system's performance during acceleration and deceleration phases. It can be stated that the measured value consistently tracks the reference speed accurately for the classical (HCC, FOC, FOC + HCC) and the proposed (FOC + HCC + SMC) controller. However, with the proposed controller, a reduction in peak overshoot (less than 1%) and a fast settling time (approximately 0.01 s) during the initial response (0–0.5 s) led to an improved speed response when compared to the classical controller. Moreover, the response time is slightly enhanced, with quick stabilization achieved within 0.005 to 0.01 s, offering faster and more efficient performance than a classical controller.

Figure 25 illustrates the zoomed trapezoidal waveform of stator back-emf for phase A. This also highlights the variation of emf concerning motor speed. The back-emf is directly proportional to the motor's rotational speed. So, when speed increases, then back-emf also increases proportionally.

Figure 26 depicts the stator current, where a similar observation can be made regarding the transient regime. By utilizing the proposed FOC + HCC + SMC controller, the peak is reduced by nearly 1.45 amperes compared to all classical controllers (Fig. 27).

Figure 28 illustrates that the integration of SMC with FOC and HCC significantly enhances torque performance by reducing ripples and improving stability. The proposed strategy achieves rapid transient stabilization (within ~0.01 s) and maintains steady-state torque with minimal deviation ($T \cong 0.2$ Nm). This makes it a robust and efficient control approach for torque regulation in BLDC motors.

Figure 29 illustrates the comparison of reference speed and the actual speed of the proposed controller. It also represents the transient and steady-state performance. The reference speed should be taken as (1000, 1200, and 800 rpm). During transitions (0.5, 1, and 1.5 s), the actual speed exhibits significant deviations from the reference speed due to sudden changes. These changes cause overshoots, where the actual speed temporarily exceeds the reference speed, followed by adjustments towards stability. The actual speed aligns closely with the reference in the steady state, with negligible steady-state error.

The key challenges faced in BLDC motors used for EVs are managing torque ripple and issues related to fault tolerance and electromagnetic interference. Controlling torque ripple is of utmost importance. This issue can be addressed effectively using control strategies such as HCC, FOC, hybrid FOC + HCC, and FOC + HCC + SMC control

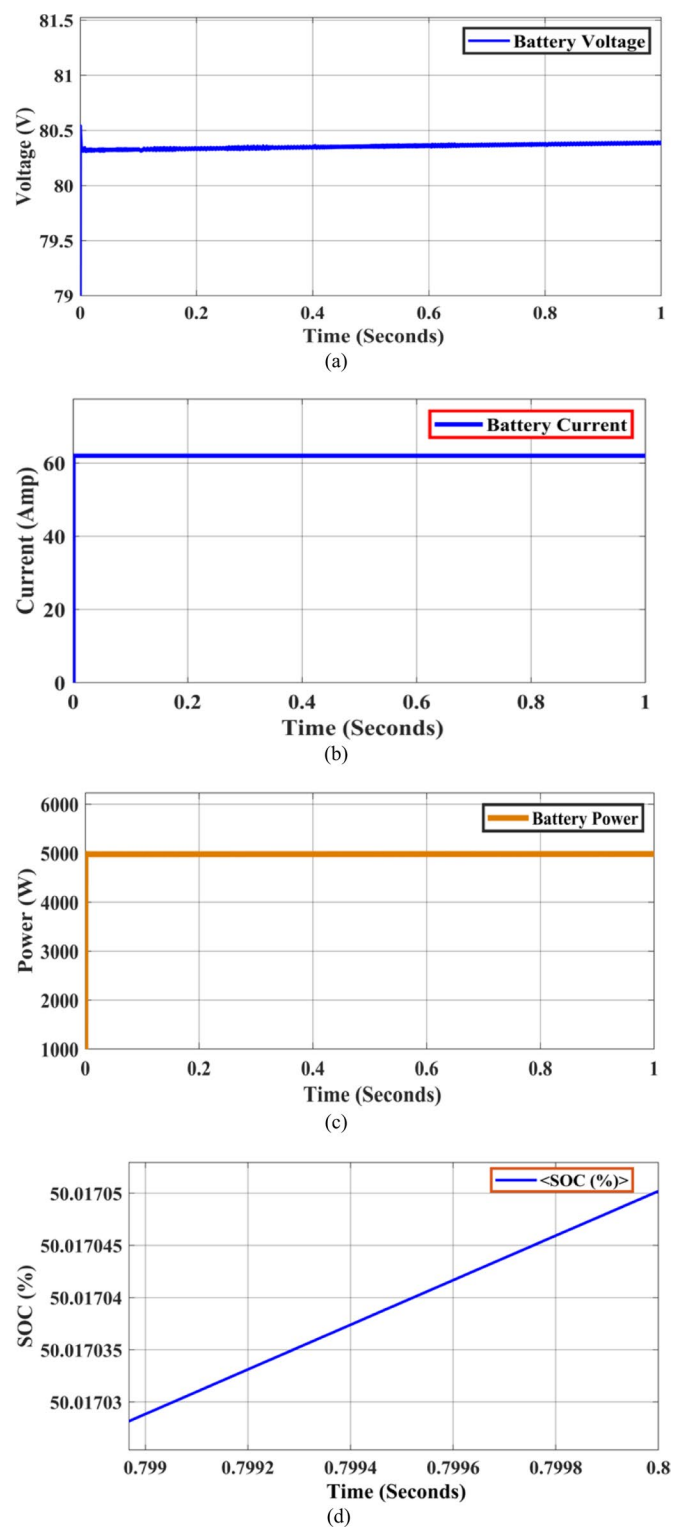


Fig. 23 **a** Battery voltage, **b** Battery current, **c** Battery power, **d** SOC (%)

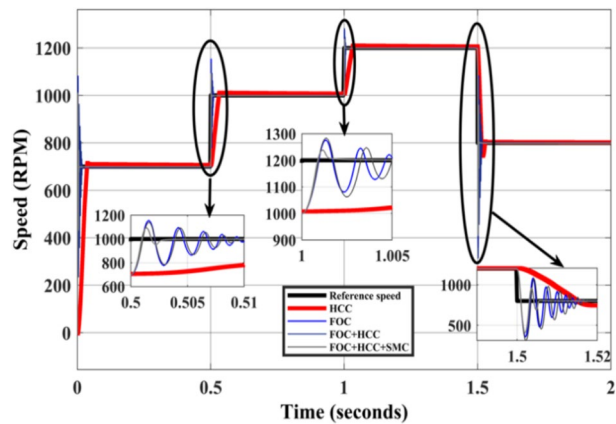


Fig. 24 An overall shape of the speed curve

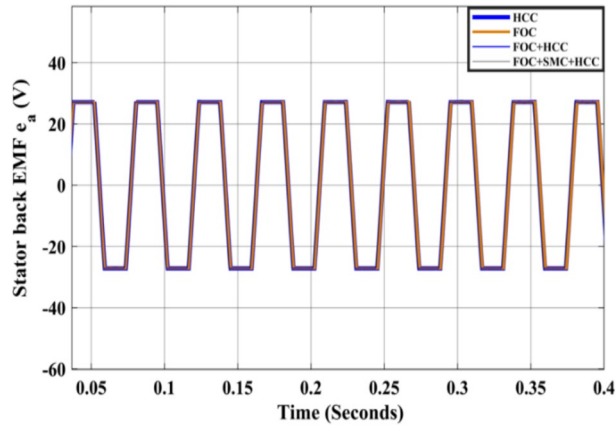


Fig. 25 Trapezoidal waveform of back electromagnetic force (EMF)

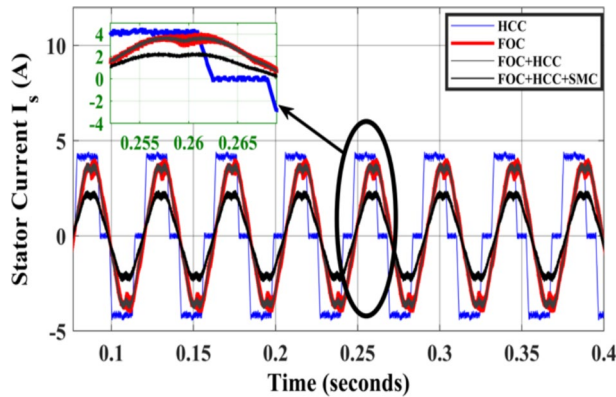


Fig. 26 Stator current **a** HCC control **b** FOC control **c** FOC+HCC control **d** Proposed (FOC+HCC+SMC) controller

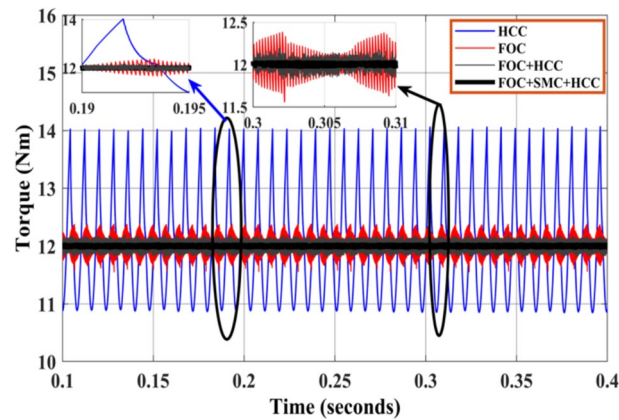


Fig. 27 An overall shape of electromagnetic torque (N-m)

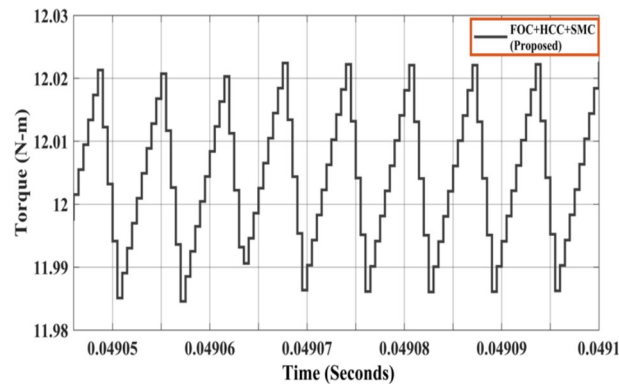


Fig. 28 Electromagnetic torque of proposed controller

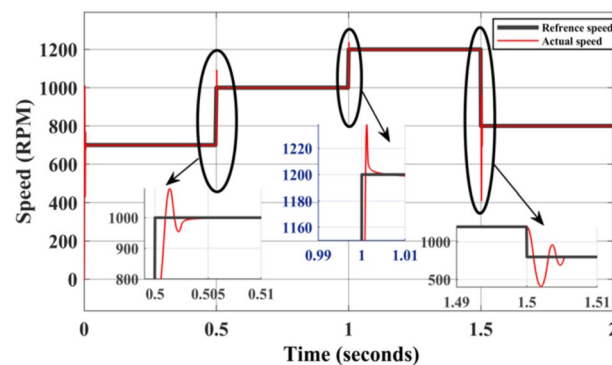


Fig. 29 Reference and actual speed of the proposed controller

methods (Fig. 30). Table 6 summarizes the unique characteristics and advantages of these techniques.

8 Conclusion

This paper comprehensively analyzes speed control strategies for BLDC motors in EV applications powered by a dual-source system integrating solar PV and grid supply. A robust MPPT-based boost converter was employed for solar energy harvesting, while a feed-forward decoupling control method ensured efficient grid power utilization.

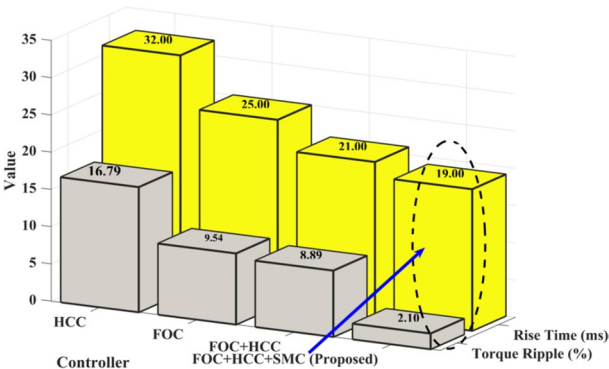


Fig. 30 Comparative analysis of all different control methods plot

Table 6 Comparative analysis of all different control methods

Controller	Torque Ripple (%)	Rise Time (msec)	Remarks
HCC	16.79	32	Simple and robust
FOC	9.54	25	Precise & efficient control with low torque ripple
FOC + HCC	8.89	21	Fast dynamic for better performance
FOC + HCC + SMC (Proposed)*	2.10	19	Highly robust and fault-tolerant with excellent torque ripple minimization

The proposed control strategies, including HCC, FOC, and hybrid techniques such as FOC + HCC and FOC + HCC + SMC, were evaluated for their effectiveness in regulating motor speed, optimizing torque performance, and enhancing overall system stability. Among these, the advanced FOC + HCC + SMC approach demonstrated superior performance by significantly reducing torque ripple, improving dynamic response, and achieving precise speed tracking under varying load and source conditions. The AI-based HQSM-AFSO optimization for MPPT provided a notable advantage by enabling faster convergence, better tracking accuracy, and higher efficiency in energy conversion compared to traditional methods. This AI-driven technique minimizes energy loss, ensuring optimal solar energy utilization even under dynamic conditions. Integrating solar and grid power ensured a reliable and stable energy supply, contributing to enhanced energy efficiency and system robustness. The FOC + HCC + SMC hybrid motor control further enhanced system performance by improving motor control precision, reducing losses, and optimizing torque distribution. The insights gained from this research advance intelligent motor control and energy management strategies, paving the way for more efficient and sustainable EV applications.

9 Future scope

We have outlined our future plan to implement a hardware-in-the-loop (HIL) experimental setup using platforms such as dSPACE or OPAL-RT. This will help to validate the proposed control strategies and power management techniques under real-time conditions and evaluate their feasibility and robustness in practical electric vehicle (EV) applications.

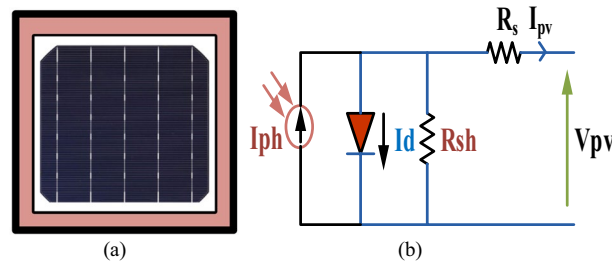


Fig. 31 a Solar PV cell. b Equivalent circuit model of PV cell

Appendix A

Specification of solar PV array

Peak power = 8 KW; open circuit (V_{oc}) voltage = 430 V; short circuit (I_{sc}) current = 18.60 A; MPP current = 8.15 A; MPP voltage = 30.7 V.

Parameters of linear transformer used in dual active bridge (DAB)

Peak power = 8 KW; V_1/V_2 (V_{rms}) = 600 V/72 V; f = 10 kHz; R_1/R_2 = 0.043/1e-6 Ω ; L_1/L_2 = 1.7e-5/1e-6 H; coupling inductor = 1.5e-6 H.

Specification of grid interface components

Grid voltage = 400 V; Grid frequency = 50 Hz; Line reactor = 0.0043H; DC-Link voltage = 600 V; DC-Link capacitance = 1000uF; PID (K_p = 20, K_d = 0 & K_i = 20); PI (K_p = 10 K_i = 10) f = 10 kHz.

Appendix B

Modeling of PV cell

The accurate representation of a PV cell is essential for optimizing energy conversion in solar systems. This study emphasized the role of the P–N junction and electron flow in power generation. Proper sunlight exposure and glass coverage significantly impact efficiency. Ensuring optimal PV cell positioning enhances performance and reliability. These insights contribute to the advancement of effective and sustainable photovoltaic systems.

From Fig. 31 we can write the equation;

$$I_{ph} = I_d + I_{Rsh} + I_{pv} \quad (44)$$

$$I_{pv} = I_{ph} - I_d - \left(\frac{V_{pv} + I_{pv}R_s}{R_{sh}} \right) \quad (45)$$

Now, the diode current, we know from the PN junction theory can be written as;

$$I_d = I_0 \left(e^{\left(\frac{V_{pv} + I_{pv}R_s}{nV_T} \right)} - 1 \right) \quad (46)$$

where V_T = Volt- equivalent of temperature = $V_T = \frac{KT}{q} = \frac{T}{11600}$, $n = 2$, for silicon, I_0 = Reverse saturation current

$$I_0 = KT^m e^{\frac{-V_{GO}}{nV_T}} \quad (47)$$

where K = constant, V_{GO} = Forbidden band gap energy, T = Temperature in Kelvin, $m = 1.5$ (Silicon), $V_{GO} = 1.16$ for Si (PV cell).

So, finally, the terminal current (I_{pv}) for the solar PV cell is given as:

$$I_{pv} = I_{ph} - I_0 \left(e^{\left(\frac{V_{pv} + I_{pv} R_s}{n V_T} \right)} - 1 \right) - \left(\frac{V_{pv} + I_{pv} R_s}{R_{sh}} \right) \quad (48)$$

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Author contributions

All listed authors have contributed to the research work presented in this manuscript. The corresponding author, Abhishek Kumar, is a postgraduate student in the Department of Electrical Engineering, MNNIT Allahabad. He was primarily responsible for carrying out the design calculations, performing simulations, and plotting the relevant graphs. Dr. M. Venkatesh Naik supervised Abhishek's MTech thesis work and contributed significantly by providing guidance on the simulation studies, design methodology, manuscript preparation, revisions, and finalization of the paper. Mr. Rahul Kumar, a Ph.D. scholar under the supervision of Dr. M. Venkatesh Naik, offered valuable inputs regarding the simulation process, design improvements, result tabulation, and graphical analysis. Aman is a postgraduate student in the Department of Electrical Engineering, MNNIT Allahabad. He was primarily responsible for carrying out the design controller. All authors have read and approved the final manuscript.

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Data availability

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Ethics approval and consent to participant

Not applicable. Accepted principles of ethical and professional conduct have been followed. No human or animal participation is involved in the research.

Consent for publication

We agree to the articles publishing terms and policies.

Competing interests

The authors declare no competing interests.

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